Predicting Building-Level Flood Risk by Analysis Comparison of Multi-Layer Perceptron and Bayesian Logistic Regression

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Introduction and Literature Review

Problem Statement

Flooding is one of the most costliest natural disasters in the US, it has causing billions of dollars in damage annually. The Federal Emergency Management Agency (FEMA) utilizes Flood Insurance Rate Maps to categorize areas by flood risk. And these categorizations, known as FIRM zones, distinguish between high-risk areas (VE, AE, A, AO, AH) and low-risk areas (Area X). However, a critical limitation of these maps is that they rely almost exclusively on macro-level topographic and hydrological data, so they are largely ignoring the specific physical characteristics of individual structures. Consequently, two houses located within the exact same flood zone can possess drastically different true vulnerabilities depending on micro-level factors such as foundation height or construction material…

Our project aims to address this critical gap. So, by leveraging detailed structural data from the National Structure Inventory (NSI), we model flood risk at the individual building level. Specifically, we implement and compare a Multilayer Perceptron (MLP) and a Bayesian Logistic Regression model to predict whether a building falls into a high-risk or low-risk flood zone. This comparison can allow us to evaluate the trade-off between the complex, non-linear pattern recognition of the MLP and the probabilistic, interpretable inference provided by the Bayesian approach, ultimately shifting the paradigm from geospatial area-mapping to structural vulnerability classification.

Refined Literature Review

The methodology we used is informed by four key prior studies. Demissie et al. (2024) has provided a foundational approach, utilize machine learning algorithms to classify flood susceptibility in Kansas based on 13 environmental features. Their models achieved high accuracy scores (0.82–0.97), identifying elevation and distance from water as the most crucial predictors. Our project adapts this rationale by substituting terrain elevation with building-specific ground_elv and found_ht features from the NSI dataset. Furthermore, Demissie et al. emphasize the importance of robust evaluation metrics, which we have incorporated into our own performance analysis.

Chen et al. (2023) directly compared a MLP against Logistic Regression and Random Forest for regional flood risk prediction. An key finding was the MLP's superior capability in capturing complex, non-linear interactions among multiple features. This was particularly

relevant to our structural analysis: a building's vulnerability is rarely determined by a single factor, but rather a complex combination of attributes (like a low foundation combined with a specific building type). Chen et al.'s findings validate our selection of an MLP, while our inclusion of Bayesian Logistic Regression provides a robust, probabilistic counterpart for baseline comparison.

Ouma and Lawrence (2023) utilizes deep learning to classify flood-prone urban environments based on broad environmental data, establishing a baseline accuracy of 82.5%. By transitioning the focus from Ouma's geographic parameter to specific structural details (such as foundation type and building age), our project hypothesizes that a building-centric model can yield even higher predictive reliability.

Finally, Qin et al. (2025) localizes flood risk assessment to the building level, analyzing how different building functions correlate with varying degrees of vulnerability within identical flood zones. Our study builds directly upon this premise but pivots the focus from building function to physical structural attributes derived from the NSI, which we argue are more direct indicators of a structure's physical resilience during a flood event.

In summary, while the existing literature successfully applies machine learning to flood risk mapping, it predominantly relies on broad geographic data or categorical building functions! This project addresses a notable gap by utilizing detailed, national-level physical building attributes (NSI) to predict FEMA flood zones, combining geospatial classification techniques with micro-level structural data.

ML Methodology and Data Collection

Data Collection and Preprocess

The dataset was sourced from the National Structure Inventory (NSI) 2022, it provides comprehensive building-level structural attributes, has including foundation type, ground elevation, building type, and structural value for all US structures. We only focused on the Massachusetts subset and it has more than 2 million datasets, and 153,239 structures of them was extracted to focus the analysis on a consistent region. The binary flood risk target was derived from the firmzone field, where any label other than AREA has been labeled as high risk, aligning with FEMA's Special Flood Hazard Area classifications. Features had been selected based on their structural and geographic relevance to flood vulnerability. All preprocess steps, including train-test splitting and standardization, was applied before the training.

Custom MLP Architecture and Implementation

To address the binary classification problems we have on identifying high-risk flood zones (VE, AE, A, etc.) based on structural building characteristics, a Custom Multi-Layer Perceptron (MLP) was been developed. Fulfilling the project requirement to implement at least one method manually without pre-built modeling packages, this neural network was constructed entirely from scratch using python and numpy for matrix operations.

The network architecture consists of a input layer corresponding to the dimensional feature space d, a single hidden layer with 16 nodes, and an output layer with a single node. The forward propagation utilizes a Rectified Linear Unit (ReLU) activation function for the hidden layer to introduce non-linearity, and a Sigmoid activation function for the output layer to map the final predictions to a probability range between 0 and 1. The mathematical formulation of the forward pass is as follows:

$$h = \max(0, XW_1 + b_1)$$

$$\hat{y} = \sigma(hW_2 + b_2) = \frac{1}{1 + e^{-(hW_2 + b_2)}}$$

The model was optimized using full-batch Gradient Descent to minimize the Binary Cross-Entropy (BCE) loss:

$$J = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

Weight matrices W^1 and W^2 were initialized using a variance-scaling strategy to prevent vanishing gradients, while biases were initialized to zero. The network was been trained over 2,000 epochs with an learning rate of η = 0.05.

Bayesian Logistic Regression Architecture and Implementation

To complement the MLP with a probabilistic approach, an Bayesian Logistic Regression (BLR) model was implemented to identify high-risk flood zones based on the same structural building characteristics. Unlike the MLP which produced point estimates, the BLR placed probability distribution over model weights, enabled uncertainty quantification for each prediction. The model was implemented using the rstanarm package in R, which uses the No-U-Turn Sampler (NUTS), an adaptive extension of Hamiltonian Monte Carlo.

The likelihood function assumed a Bernoulli distribution for the binary flood risk outcome with the probability parameterized through the logistic sigmoid function:

$$y_i | W, x_i \sim Bern(\theta = \sigma(W^T x_i))$$

$$\sigma(W^T x_i) = 1/\left(1 + e^{(-W^T x_i)}\right)$$

Weak informative priors were placed over all model weights to regularize the posterior while allowing the data to make changes. Features' weights were assigned a Normal prior centered at zero with a standard deviation of 2.5, while the intercept was given a wider prior to allow the data to adjust freely:

$$W_j \sim N(0, 0.25^2)\ for\ j = 1, \ldots, 15$$

$$W_0 \sim N(0, 10^2)$$

The posterior distribution over all 16 weights were computed by Bayes' theorem:

$$p(W|X, y) \propto p(y|W, X) \cdot p(W)$$

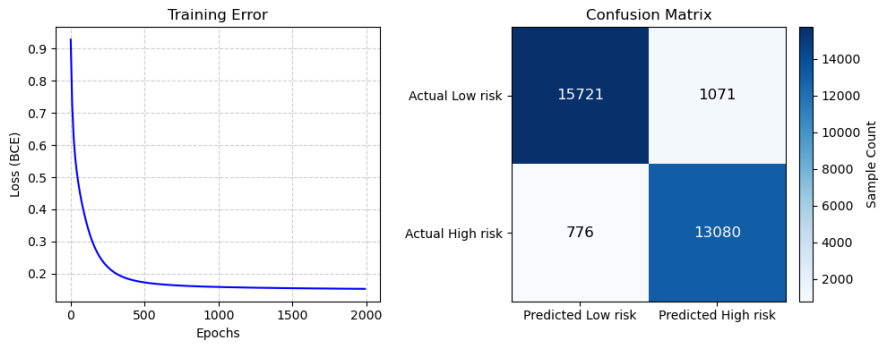
Posterior inference was conducted using 4 independent MCMC chains, each running 1000 iterations with 500 warmup iterations discarded, yielding 2000 posterior draws. Predictions was obtained by averaging the posterior predictive probabilities across all 2000 draws, with observations classified as high risk when the mean predicted probability exceeded 0.5.

ML Results and Interpretation

MLP Predictive Performance

The custom MLP model demonstrated exceptionally strong predictive capabilities on the unseen test set. The model achieved an overall Accuracy of 93.97%, indicated a high degree of reliability in classifying building vulnerability.

Further evaluation using the Confusion Matrix (figure below, right) reveals a robust balance between sensitivity and precision. The model achieved a Precision of 0.9243 and a Recall of 0.9440, culminating in an F1-Score of 0.9341. In the context of predicting high-risk zones, the high Recall is particularly critical; it indicates a very low rate of False Negatives. Minimizing False Negatives ensures that vulnerable structures are not mistakenly classified as safe, which is paramount for effective municipal planning and risk assessment.



Additionally, the Training Error curve (see figure above, left) illustrates a smooth convergence of the Binary Cross-Entropy loss, dropped very obviously within the first 200 epochs and stabilizing around 0.1533 by the 2000th epoch, confirming that the manual gradient descent implementation effectively optimized the weight parameters without overfitting.

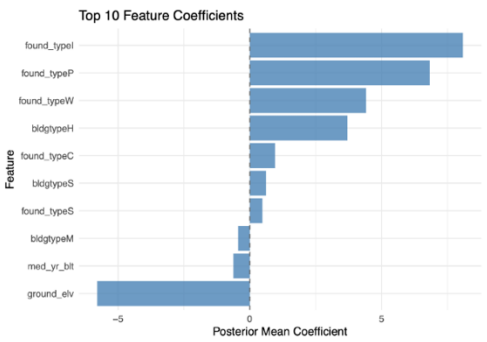
Bayesian Logistic Regression Predictive Performance

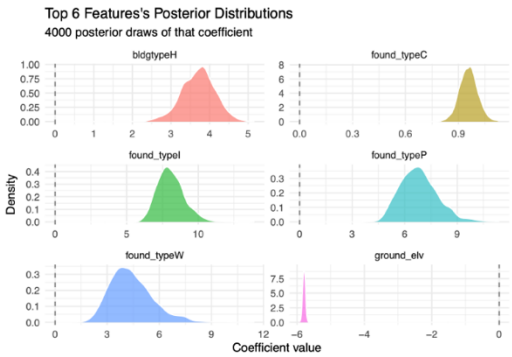
The Bayesian Logistic Regression model demonstrates strong predictive capabilities on the unseen test set. The model achieves an overall Accuracy of 92.52%, reflecting a high reliability in classifying building vulnerability across the Massachusetts region.

Further evaluation by using the Confusion Matrix has reveals a favorable balance between recall and precision. The model achieved a Precision of 0.8890 and Recall of 0.7675, culminating in an F1-Score of 0.8238. While the Recall is lower than that of the MLP, the

Precision of 0.8890 indicates that when the model predicts a structure as high risk, it is correct nearly 89% of the time. In the context of municipal flood risk planning, this ensured that resources directed toward flagged structures are well target and not distracted by excessive false positives.

A key advantage of the Bayesian Logistic Regression over the MLP is its interpretability through posterior coefficient estimates. Foundation type emerges as the dominant driver of flood risk classification. Pile foundations (found_typeI) and pier foundations (found_typeP) exhibit the strongest positive associations. This is very intuitive since they are the main structures located in coastal and riverine flood zones. Ground elevations (ground_elv) exhibits the strongest negative coefficient across all features, confirmed that higher elevation is the most robust protective factor against flood risk designation.





Conclusions and Future Work

MLP: Drawbacks and Model Limitations

While the custom MLP achieved high predictive accuracy(93.97%), a primary drawback of this approach is its "black-box" nature. So unlike traditional linear or logistic regression models, neural networks does not provide easily interpretable coefficients. Consequently, we can’t directly extract feature importance to determine exactly which structural variables (like ground elevation, foundation height, or building type) act as the primary drivers for flood risk classification…

If this problem were to be expanded in future work, incorporating a post-hoc interpretability framework—such as SHAP (SHapley Additive exPlanations) values—would be highly beneficial to understand the marginal contribution of individual building attributes to the model's predictions!

BLR: Findings and Drawbacks

The Bayesian Logistic Regression achieved 92.52% accuracy while provide interpretable posterior coefficient estimates. Also, confirming that foundation type and ground elevation are the primary structural determinants of flood risk in Massachusetts. However, the model limits its ability to capture complex non-linear interactions, which lower Recall of 0.7675

compared to the MLP.

Model Assumptions and Data Appropriateness

Regarding our model assumptions, BLR has fundamentally assumes a linear relationship between the independent variables and the log-odds of the high-risk flood designation. The fact that we could see is, the MLP outperform the BLR, particularly in Recall (0.9440 vs 0.7675), which suggests that this linearity assumption may not perfectly hold for our dataset, as true structural vulnerability is likely driven by complex, non-linear interactions among features. Additionally, BLR assumes the independence of observations; while our data represents distinct structures, some spatial autocorrelation might exist in real-world flood zones (buildings in close proximity sharing similar localized risks). Conversely, the MLP does not rely on these strict statistical assumption, made it highly appropriate for mapping non-linear structural patterns. The primary requirement for the MLP was that the input space be properly scaled to ensure stable gradient descent convergence, which was successfully met through our standardization step and validate by the smooth convergence of training errors.

Future work

For future work, the implementation of a Bayesian Neural Network (BNN) would address the primary limitations of both models by combining the predictive capacity of deep neural networks with uncertainty quantification. A BNN could potentially achieve higher recall on high-risk structures without sacrificing the uncertainty estimation of probabilistic models for municipal flood risk planning. Incorporating spatial features such as coastlines and elevation relative to base flood elevation would also likely improve predictive performance across both approaches.

Appendix: Implementation

MLP:

```
def sigmoid(z):

    z = np.clip(z, -500, 500)

    return 1 / (1 + np.exp(-z))



n = X_train.shape[0]

d = X_train.shape[1]

hidden_nodes = 16

out_nodes = 1
```

```
eta = 0.05

epochs = 2000


np.random.seed(33)

W1 = np.random.randn(d, hidden_nodes) * np.sqrt(2.0 / d)

b1 = np.zeros((1, hidden_nodes))

W2 = np.random.randn(hidden_nodes, out_nodes) * np.sqrt(1.0 / hidden_nodes)

b2 = np.zeros((1, out_nodes))

def f(x):

    h = np.maximum(0, x.dot(W1) + b1)

    return sigmoid(h.dot(W2) + b2)


def compute_bce_loss(Y_true, Y_pred):

    epsilon = 1e-8

    Y_pred = np.clip(Y_pred, epsilon, 1 - epsilon)

      return -np.mean(Y_true * np.log(Y_pred) + (1 - Y_true) * np.log(1 - Y_pred))


errors = []


for epoch in range(epochs):

    h = np.maximum(0, X_train.dot(W1) + b1)

    y_pred = sigmoid(h.dot(W2) + b2)

    dy = y_pred - Y_train

    dW2 = h.T.dot(dy) / n

    db2 = np.sum(dy, axis=0, keepdims=True) / n

    mat1 = np.heaviside(h, 0)

    dh = dy.dot(W2.T)


    dW1 = X_train.T.dot(dh * mat1) / n

    db1 = np.sum(dh * mat1, axis=0, keepdims=True) / n
```

```
    W2 = W2 - eta * dW2

    b2 = b2 - eta * db2

    W1 = W1 - eta * dW1

    b1 = b1 - eta * db1


    if epoch % 10 == 0:

        e = compute_bce_loss(Y_train, y_pred)

        errors.append(e)

        if epoch % 200 == 0:

            print(f"Epoch {epoch:<4} | Loss: {e:.4f}")
```

BLR In R:

```
blr_model <- stan_glm(

  target_label ~ .,

  data = train_df,

  family = binomial(link = "logit"),

  prior = normal(0, 2.5),

  prior_intercept = normal(0, 10),

  chains = 4,

  iter = 1000,

  warmup = 500,

  cores = 4,

  seed = 33,

  refresh = 200

)
```

Repo: https://github.com/samueeelsiu/DS4420-Final-Proj